

Finite-Time and Asymptotic Convergence of Gradient-Flow Methods for Bilevel Optimization

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Abstract—Bilevel optimization problems, in which an upper-level objective is minimized subject to the optimality of a lower-level variable, are ubiquitous in meta-learning and control. While discrete-time algorithms are well-studied, the continuous-time properties of these problems offer deeper theoretical insights into stability and convergence rates. This paper investigates gradient-flow dynamics for solving bilevel optimization problems across three distinct regularity regimes. We propose a dynamical system based on a hypergradient estimator and analyze its convergence using Lyapunov stability theory. Specifically, we introduce a normalized gradient flow that achieves *finite-time convergence* to the global optimum when both levels are strongly convex. Furthermore, we relax these assumptions to Polyak-Łojasiewicz (PL) conditions, demonstrating convergence to a residual neighborhood of the optimum for PL upper-level objectives, and asymptotic convergence for non-convex upper-level objectives. We provide rigorous derivations of the hypergradient error bounds and validate the theoretical results via numerical simulations on synthetic problems.

I. INTRODUCTION

Bilevel optimization provides a mathematical framework for hierarchical decision problems in which an upper-level objective depends on the solution of a lower-level optimization. These formulations are central to hyperparameter optimization and meta-learning [1], Stackelberg games [2], and model predictive control [3]. In the canonical formulation the outer agent minimizes $F(x)$ while the inner agent returns a best-response $y^*(x)$, yielding an implicit dependence between the two optimization problems (see formulation (1)).

A central computational challenge is estimating the hypergradient—the derivative of the upper objective that propagates through the lower-level solution map. Existing approaches trade off cost, memory, and assumptions: nested (double-loop) schemes that repeatedly solve the lower problem [4], implicit-differentiation methods that invert the lower-level Hessian via the implicit function theorem [5], and differentiate-through-optimization methods that back-propagate through the solver trajectory [6]. Classical single-level reformulations add further difficulty: recasting the lower-level KKT conditions as a mathematical program with equilibrium constraints (MPEC) is not equivalent in general [7] and demands delicate constraint qualifications, while a nonconvex lower level renders the best-response map set-valued, forcing optimistic/pessimistic formulations and explicit selection rules [8]. Consequently, practical methods either impose strong structural assumptions (strong convexity, Hessian invertibility, unique minimizers) to justify

implicit differentiation [4], or resort to heuristic truncation with weaker guarantees [6].

Casting bilevel optimality conditions as a coupled dynamical system and analyzing them via Lyapunov methods offers a complementary, more transparent lens: it exposes the dynamical structure, yields stability guarantees without repeated inner solves, and under suitable conditions admits finite-time convergence, as in related fixed-time frameworks [9]. Control-theoretic gradient flows for bilevel problems illustrate how continuous-time analysis both clarifies behavior and guides the design of practical discrete algorithms [8].

Several recent works advance bilevel theory and algorithms. Penalty-based bilevel gradient descent methods for nonconvex upper-level objectives have been proposed in [10], while a single-timescale stochastic method without nested loops was developed in [11]. For simple bilevel programs with lower-level objectives independent of x , continuous-time diagonal dynamics were analyzed using Lyapunov techniques in [12], while the GALET framework in [13] addresses problems with a PL lower level objective. Closest to present work, [14] analyzes a continuous-time flow for non-convex upper / strongly-convex lower problems, obtaining $O(1/k)$ ergodic convergence. In contrast, our analysis covers three regularity regimes and establish *finite-time pointwise* convergence under joint strong convexity.

Our contributions are threefold. **(i) Finite-time convergence:** a normalized gradient flow that provably reaches the optimum in finite time when both levels are strongly convex. To the best of our knowledge, a novel result for bilevel optimization dynamics. **(ii) PL relaxation:** under a PL upper level, the dynamics converge to a bounded neighborhood whose size is set by the problem conditioning. **(iii) Non-convex upper level:** asymptotic convergence guarantees via the standard gradient flow.

Notation: Throughout, $x \in \mathbb{R}^{d_x}$ and $y \in \mathbb{R}^{d_y}$ are the upper and lower-level variables. $y^*(x)$ is the lower-level solution, defining $F(x) := f(x, y^*(x))$ with optimal value F^* . Strong convexity parameters for $g(x, \cdot)$, $f(x, y)$, $F(x)$ are μ_g, μ_f, μ_F . Lipschitz constants for $\nabla f, \nabla g, \nabla_y f$ are L_f, L_g, L_{fy} . Hessian Lipschitz constants are L_{gxy}, L_{gyy} for $\nabla_{xy}^2 g, \nabla_{yy}^2 g$. Uniform bounds: $C_f := \sup \|\nabla_y f\|$ and $C_{gxy} := \sup \|\nabla_{xy}^2 g\|$. The Lipschitz constant of $y^*(x)$ is $L_y := C_{gxy}/\mu_g$, and C_M bounds $\|\partial y^*(x)/\partial x\|$. The hypergradient estimator is $\mathcal{V}(x, y)$ with error constant L_v . $\alpha, \beta > 0$ are the upper and lower-level step sizes.

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II. PRELIMINARIES

We study the bilevel optimization problem of the form:

$$\begin{aligned} \min_{x \in \mathbb{R}^{d_x}} F(x) &= f(x, y^*(x)), \\ y^*(x) &:= \arg \min_{y \in \mathbb{R}^{d_y}} g(x, y) \end{aligned} \quad (1)$$

under suitable assumptions on function regularity and convexity. Below we state these assumptions (some vary by case, but we maintain a global naming scheme and note which ones apply).

Assumption 1 (Lower-level strong convexity). *For each $x \in \mathbb{R}^{d_x}$, the function $y \mapsto g(x, y)$ is μ_g -strongly convex:*

$$(\nabla_y g(x, y_1) - \nabla_y g(x, y_2))^\top (y_1 - y_2) \geq \mu_g \|y_1 - y_2\|^2.$$

In particular, for each x the lower-level minimizer $y^(x)$ is unique.*

Assumption 2 (Upper-level strong convexity). *The function $(x, y) \mapsto f(x, y)$ is μ_f -strongly convex:*

$$(\nabla f(z_1) - \nabla f(z_2))^\top (z_1 - z_2) \geq \mu_f \|z_1 - z_2\|^2,$$

for all $z_i = [x_i, y_i]$, $\mu_f > 0$.

Assumption 3 (Upper-level PL). *The function $f(x, y)$ satisfies the PL inequality:*

$$\frac{1}{2} \|\nabla f(x, y)\|^2 \geq \mu_f (f(x, y) - f^*), \quad \forall (x, y),$$

where $f^* = \inf_{x, y} f(x, y)$, $\mu_f > 0$.

Assumption 4 (Smoothness and Hessian Lipschitz). *f, g are twice continuously differentiable, and there exist constants $L_f, L_g, L_{gxy}, L_{gyy} > 0$ such that, for all (x_i, y_i) , each of $\nabla f, \nabla g, \nabla_{xy}^2 g, \nabla_{yy}^2 g$ is Lipschitz with the corresponding constant; e.g.,*

$$\|\nabla f(x_1, y_1) - \nabla f(x_2, y_2)\| \leq L_f (\|x_1 - x_2\| + \|y_1 - y_2\|),$$

and analogously for ∇g (constant L_g) and for the Hessians $\nabla_{xy}^2 g, \nabla_{yy}^2 g$ (constants L_{gxy}, L_{gyy}).

Assumption 5 (Bounded gradients and Hessians). *There exist positive constants C_f, C_{gxy} such that for all x :*

$$\|\nabla_y f(x, y)\| \leq C_f, \quad \|\nabla_{xy}^2 g(x, y)\| \leq C_{gxy}.$$

Assumption 6 (Boundedness of objective). *$F(x) = f(x, y^*(x))$ is bounded below: $F^* := \inf_x F(x) > -\infty$.*

A. Lipschitz continuity of $y^(x)$*

Lemma 1 (Lipschitz continuity of $y^*(x)$). *Under Assumptions 1, 4, and 5 we have*

$$\|y^*(x_1) - y^*(x_2)\| \leq L_y \|x_1 - x_2\|, \quad L_y = \frac{C_{gxy}}{\mu_g}.$$

Proof. Differentiating $\nabla_y g(x, y^*(x)) = 0$ gives $\partial y^*(x)/\partial x = -[\nabla_{yy}^2 g]^{-1} \nabla_{xy}^2 g$ (evaluated at $(x, y^*(x))$). Taking norms and using $\|[\nabla_{yy}^2 g]^{-1}\| \leq \mu_g^{-1}$ (Assumption 1) and $\|\nabla_{xy}^2 g\| \leq C_{gxy}$ (Assumption 5) yields $\|\partial y^*(x)/\partial x\| \leq C_{gxy}/\mu_g = C_M$. The bound

$\|y^*(x_1) - y^*(x_2)\| \leq L_y \|x_1 - x_2\|$ then follows from the mean value theorem. $\square$

Lemma 2 (Strong convexity of composed objective F). *Assume f is μ_f -strongly convex (Assumption 2), and $\nabla_y f$ is Lipschitz in (x, y) with constant L_{fy} and is uniformly bounded by C_f . Let $M(x) = \partial y^*(x)/\partial x$ with $\|M(x)\| \leq C_M$. Then $F(x) = f(x, y^*(x))$ is μ_F -strongly convex with*

$$\mu_F = \mu_f - [L_{fy}(1 + L_y)L_y + 2C_f C_M]. \quad (2)$$

In particular, if $\mu_f > L_{fy}(1 + L_y)L_y + 2C_f C_M$, then $\mu_F > 0$ and F is strongly convex.

Proof. Write $z_i = [x_i, y_i^*]$ with $y_i^* := y^*(x_i)$, $v := x_1 - x_2$, $M_i := M(x_i)$, and $h_i := \nabla_y f(z_i)$. By the chain rule $\nabla F(x) = \nabla_x f(x, y^*(x)) + M(x)^\top \nabla_y f(x, y^*(x))$, so

$$\begin{aligned} \langle \nabla F(x_1) - \nabla F(x_2), v \rangle &= \langle \nabla_x f(z_1) - \nabla_x f(z_2), v \rangle \\ &\quad + \langle M_1^\top h_1 - M_2^\top h_2, v \rangle. \end{aligned}$$

The μ_f -strong convexity of f in (x, y) reads $\langle \nabla f(z_1) - \nabla f(z_2), z_1 - z_2 \rangle \geq \mu_f (\|v\|^2 + \|y_1^* - y_2^*\|^2)$; isolating the x -block gives

$$\begin{aligned} \langle \nabla_x f(z_1) - \nabla_x f(z_2), v \rangle &\geq \mu_f \|v\|^2 + \mu_f \|y_1^* - y_2^*\|^2 \\ &\quad - \langle h_1 - h_2, y_1^* - y_2^* \rangle. \end{aligned}$$

Substituting and collecting the cross terms into

$$R := -\langle h_1 - h_2, y_1^* - y_2^* \rangle + \langle h_1, M_1 v \rangle - \langle h_2, M_2 v \rangle,$$

we obtain $\langle \nabla F(x_1) - \nabla F(x_2), v \rangle \geq \mu_f \|v\|^2 + \mu_f \|y_1^* - y_2^*\|^2 + R$. By Cauchy-Schwarz with $\|h_1 - h_2\| \leq L_{fy}(1 + L_y)\|v\|$, $\|y_1^* - y_2^*\| \leq L_y\|v\|$, $\|h_i\| \leq C_f$, and $\|M_i\| \leq C_M$,

$$|R| \leq [L_{fy}(1 + L_y)L_y + 2C_f C_M]\|v\|^2.$$

Dropping the nonnegative term $\mu_f \|y_1^* - y_2^*\|^2$ and using $R \geq -|R|$,

$$\begin{aligned} \langle \nabla F(x_1) - \nabla F(x_2), v \rangle &\geq (\mu_f - [L_{fy}(1 + L_y)L_y + 2C_f C_M])\|v\|^2, \end{aligned}$$

which is exactly $\mu_F \|v\|^2$ with μ_F as in (2). $\square$

III. KEY RESULTS

Solving the bilevel problem relies on defining the hypergradient estimator $\mathcal{V}(x, y)$ as a proxy for the true hypergradient $\nabla F(x)$. Following [4], we define:

$$\begin{aligned} \mathcal{V}(x, y) &= \nabla_x f(x, y) \\ &\quad - \nabla_{xy}^2 g(x, y) [\nabla_{yy}^2 g(x, y)]^{-1} \nabla_y f(x, y). \end{aligned} \quad (3)$$

Note that when $y = y^*(x)$, this estimator coincides with the true gradient: $\mathcal{V}(x, y^*(x)) = \nabla F(x)$.

We now present derivations for the hypergradient error bounds and quadratic-growth used repeatedly.

Lemma 3 (Hypergradient error bound). *Under Assumptions 1, 4, and 5, there exists $L_v > 0$ such that*

$$\|\mathcal{V}(x, y) - \nabla F(x)\| \leq L_v \|y - y^*(x)\|,$$

$$L_v = L_f + \frac{L_{gxy}C_f}{\mu_g} + \frac{C_{gxy}L_{ggy}C_f}{\mu_g^2} + \frac{C_{gxy}L_f}{\mu_g}. \quad (4)$$

The estimate follows the analysis of [4]; we include the derivation for completeness.

Proof. Computing the difference:

$$\begin{aligned} \mathcal{V}(x, y) - \nabla F(x) &= [\nabla_x f(x, y) - \nabla_x f(x, y^*(x))] \\ &\quad - \left(\nabla_{xy}^2 g(x, y) [\nabla_{yy}^2 g(x, y)]^{-1} \nabla_y f(x, y) \right. \\ &\quad \left. - \nabla_{xy}^2 g(x, y^*) [\nabla_{yy}^2 g(x, y^*)]^{-1} \nabla_y f(x, y^*) \right). \end{aligned}$$

Decomposing the second term using $BC - B'C' = (B - B')C + B'(C - C')$ and a second decomposition on $(C - C')$ for the product of three matrices:

$$\begin{aligned} &\nabla_{xy}^2 g(x, y) [\nabla_{yy}^2 g(x, y)]^{-1} \nabla_y f(x, y) \\ &\quad - \nabla_{xy}^2 g(x, y^*) [\nabla_{yy}^2 g(x, y^*)]^{-1} \nabla_y f(x, y^*) \\ &= [\nabla_{xy}^2 g(x, y) - \nabla_{xy}^2 g(x, y^*)] [\nabla_{yy}^2 g(x, y)]^{-1} \nabla_y f(x, y) \\ &\quad + \nabla_{xy}^2 g(x, y^*) \{ [\nabla_{yy}^2 g(x, y)]^{-1} - [\nabla_{yy}^2 g(x, y^*)]^{-1} \} \\ &\quad \times \nabla_y f(x, y) \\ &\quad + \nabla_{xy}^2 g(x, y^*) [\nabla_{yy}^2 g(x, y^*)]^{-1} \\ &\quad \times [\nabla_y f(x, y) - \nabla_y f(x, y^*)]. \end{aligned}$$

We bound each of the three terms using Hessian Lipschitz bounds and strong convexity. *First:* $\|\nabla_{xy}^2 g(x, y) - \nabla_{xy}^2 g(x, y^*)\| \leq L_{gxy}\|y - y^*\|$, $\|[\nabla_{yy}^2 g]^{-1}\| \leq \mu_g^{-1}$, $\|\nabla_y f\| \leq C_f$, giving $\frac{L_{gxy}C_f}{\mu_g}\|y - y^*\|$. *Second:* using the identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ together with $\|\nabla_{xy}^2 g\| \leq C_{gxy}$, $\|[\nabla_{yy}^2 g]^{-1}\| \leq \mu_g^{-1}$, and the Hessian-Lipschitz bound $\|\nabla_{xy}^2 g(x, y) - \nabla_{xy}^2 g(x, y^*)\| \leq L_{gxy}\|y - y^*\|$, this term is at most $\frac{C_{gxy}L_{ggy}C_f}{\mu_g^2}\|y - y^*\|$. *Third:* $\|\nabla_y f(x, y) - \nabla_y f(x, y^*)\| \leq L_f\|y - y^*\|$ and $\|[\nabla_{yy}^2 g]^{-1}\| \leq \mu_g^{-1}$ give $\frac{C_{gxy}L_f}{\mu_g}\|y - y^*\|$.

The direct $\nabla_x f$ difference is bounded as $\|\nabla_x f(x, y) - \nabla_x f(x, y^*)\| \leq L_f\|y - y^*(x)\|$.

Combining all terms yields the stated inequality with L_v given by (4). $\square$

Lemma 4 (Quadratic-growth and PL consequences). *Under lower-level $g(x, \cdot)$ strong convexity (Assumption 1), we have*

$$\begin{aligned} \|y - y^*(x)\|^2 &\leq \frac{2}{\mu_g} (g(x, y) - g(x, y^*(x))), \\ \|\nabla_y g(x, y)\|^2 &\geq 2\mu_g (g(x, y) - g(x, y^*(x))). \end{aligned}$$

If f is μ_f -strongly convex, then by Lemma 2 under

$$0 < \mu_f - [L_{fy}(1 + L_y)L_y + 2C_fC_M],$$

we can state

$$\|\nabla F(x)\|^2 \geq 2\mu_F(F(x) - F^*).$$

Proof. The first two inequalities are the standard quadratic-growth and gradient-domination consequences of μ_g -strong convexity; the third is the PL inequality for F , which follows from its μ_F -strong convexity (Lemma 2). See [15] for these classical equivalences. $\square$

IV. DYNAMICAL SYSTEMS AND CONVERGENCE ANALYSIS

In this section, we propose continuous-time dynamical systems to solve the bilevel optimization problem (1). To facilitate the analysis, we introduce two types of gradient flows:

(a) *Normalized gradient flow (used in Cases B and C):*

$$\dot{x} = -\alpha \frac{\mathcal{V}(x, y)}{\|\mathcal{V}(x, y)\|}, \quad \dot{y} = -\beta \frac{\nabla_y g(x, y)}{\|\nabla_y g(x, y)\|} \quad (5)$$

with $\alpha, \beta > 0$.

(b) *Standard gradient flow (used in Case A):*

$$\dot{x} = -\alpha \mathcal{V}(x, y), \quad \dot{y} = -\beta \nabla_y g(x, y) \quad (6)$$

with $\alpha, \beta > 0$.

We analyze the stability and convergence across three distinct regularity regimes for the upper-level objective $f(x, y)$.

A. Case A: $f(x, y)$ non-convex, $g(x, \cdot)$ μ_g -strongly convex

Assumptions: 4, 5, 6; no regularity assumed on $f(x, y)$.

Dynamics: Standard gradient flow (6) with estimator $\mathcal{V}$.

Lyapunov functions:

$$\begin{aligned} V_1(x) &= F(x) - F^*, \\ V_2(x, y) &= g(x, y) - g(x, y^*(x)), \\ V(x, y) &= V_1(x, y) + V_2(x, y). \end{aligned} \quad (7)$$

Lemma 5 (Time derivative of V_1). *Consider the dynamics (6). Under Assumptions 4 and 5, and utilizing the error bound L_v derived in Lemma 3, the time derivative of the upper-level value function satisfies:*

$$\dot{V}_1 \leq -\frac{\alpha}{2} \|\nabla F(x)\|^2 + \frac{\alpha L_v^2}{2\mu_g^2} \|\nabla_y g(x, y)\|^2.$$

Proof. Since $\dot{x} = -\alpha \mathcal{V}(x, y)$,

$$\begin{aligned} \dot{V}_1 &= \langle \nabla F(x), \dot{x} \rangle = -\alpha \|\nabla F(x)\|^2 \\ &\quad + \alpha \langle \nabla F(x), \nabla F(x) - \mathcal{V}(x, y) \rangle. \end{aligned}$$

By Cauchy-Schwarz with Lemma 3 and Young's inequality $ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2$,

$$\begin{aligned} \dot{V}_1 &\leq -\alpha \|\nabla F(x)\|^2 + \alpha L_v \|\nabla F(x)\| \|y - y^*(x)\| \\ &\leq -\frac{\alpha}{2} \|\nabla F(x)\|^2 + \frac{\alpha L_v^2}{2} \|y - y^*(x)\|^2. \end{aligned}$$

Invoking Lemma 4, $\|y - y^*(x)\|^2 \leq \mu_g^{-2} \|\nabla_y g(x, y)\|^2$, which yields the stated inequality. $\square$

Lemma 6 (Time derivative of V_2). *Consider the dynamics (6). For any $\gamma > 0$, the time derivative of the lower-level value function satisfies:*

$$\dot{V}_2 \leq \frac{\alpha^2 L_g^2}{2\gamma} \|\mathcal{V}(x, y)\|^2 + \left(\frac{\gamma}{2\mu_g^2} - \beta \right) \|\nabla_y g(x, y)\|^2.$$

Proof. Differentiating along (6) with $\dot{y} = -\beta \nabla_y g(x, y)$,

$$\dot{V}_2 = \langle \nabla_x g(x, y) - \nabla_x g(x, y^*(x)), \dot{x} \rangle - \beta \|\nabla_y g(x, y)\|^2.$$

Since $\dot{x} = -\alpha \mathcal{V}(x, y)$ and $\nabla_x g$ is L_g -Lipschitz in its second argument (Assumption 4), the first term is at most

$\alpha L_g \|y - y^*(x)\| \|\mathcal{V}(x, y)\|$. Applying Young's inequality $ab \leq \frac{a^2}{2\gamma} + \frac{\gamma b^2}{2}$ (with $a = \alpha L_g \|\mathcal{V}(x, y)\|$, $b = \|y - y^*(x)\|$) followed by the quadratic-growth bound $\|y - y^*(x)\|^2 \leq \mu_g^{-2} \|\nabla_y g(x, y)\|^2$ (Lemma 4) yields the claimed inequality. $\square$

1) Convergence Analysis: By combining the descent estimates in Lemmas 5 and 6, we establish asymptotic convergence.

Theorem 1 (Asymptotic convergence). *Consider the bi-level problem (1) where $F(x)$ is bounded below (Assumption 6) and f, g satisfy Assumptions 4 and 5. Let the system evolve according to the Standard Gradient Flow (6). There exist parameters $\alpha, \beta > 0$ and an auxiliary constant $\gamma > 0$ such that:*

$$\lim_{t \rightarrow \infty} \|\nabla F(x(t))\| = 0, \quad \lim_{t \rightarrow \infty} \|\nabla_y g(x(t), y(t))\| = 0.$$

Proof. Combining Lemmas 5 and 6 with $V = V_1 + V_2$:

$$\begin{aligned} \dot{V} \leq & -\frac{\alpha}{2} \|\nabla F(x)\|^2 + \frac{\alpha L_v^2}{2\mu_g^2} \|\nabla_y g\|^2 \\ & + \frac{\alpha^2 L_g^2}{2\gamma} \|\mathcal{V}(x, y)\|^2 + \left(\frac{\gamma}{2\mu_g^2} - \beta\right) \|\nabla_y g\|^2. \end{aligned}$$

Bounding $\|\mathcal{V}(x, y)\|^2 \leq 2\|\nabla F(x)\|^2 + 2L_v^2 \|y - y^*(x)\|^2 \leq 2\|\nabla F(x)\|^2 + \frac{2L_v^2}{\mu_g^2} \|\nabla_y g\|^2$:

$$\dot{V} \leq \left(-\frac{\alpha}{2} + \frac{\alpha^2 L_g^2}{\gamma}\right) \|\nabla F(x)\|^2 + A(\gamma, \alpha, \beta) \|\nabla_y g\|^2,$$

where

$$A(\alpha, \beta, \gamma) = \frac{\alpha L_v^2}{2\mu_g^2} + \frac{\alpha^2 L_g^2 L_v^2}{\gamma \mu_g^2} + \frac{\gamma}{2\mu_g^2} - \beta.$$

Choosing parameters $\alpha, \beta, \gamma > 0$ such that

$$-\frac{\alpha}{2} + \frac{\alpha^2 L_g^2}{\gamma} < 0, \quad A(\alpha, \beta, \gamma) < 0,$$

the Lyapunov function satisfies

$$\dot{V} \leq -c_1 \|\nabla F(x)\|^2 - c_2 \|\nabla_y g(x, y)\|^2, \quad c_1, c_2 > 0.$$

Hence $\dot{V} \leq 0$, and V is non-increasing. By LaSalle's invariance principle, every trajectory converges to the largest invariant set in

$$\{(x, y) : \dot{V} = 0\} = \{(x, y) : \nabla_y g(x, y) = 0\},$$

which by strong convexity of $g(x, \cdot)$ is precisely $\{(x, y) : y = y^*(x)\}$. On this manifold, $\dot{x} = -\alpha \nabla F(x)$, so the limit set satisfies $\nabla F(x) = 0$. Consequently, all trajectories converge to $\{(x, y) : y = y^*(x), \nabla F(x) = 0\}$. $\square$

B. Case B: $f(x, y)$ μ_f -strongly convex and $g(x, \cdot)$ μ_g -strongly convex

Given $\mu_f = \mu_F + L_{fy}(1 + L_y)L_y + 2C_f C_M$, we use normalized flow (5) for finite-time convergence. *Assumptions:* 1, 2, 4, 5, 6. *Dynamics:* Normalized flow (5) with $\mathcal{V}$ from (3). *Lyapunov:* V_1, V_2, V from (7).

Lemma 7 (Time derivative of V_1). *Under the normalized dynamics (5), the upper-level Lyapunov candidate satisfies:*

$$\dot{V}_1 \leq -\alpha \left(\frac{\|\nabla F(x)\|}{2} - \frac{L_v \|y - y^*(x)\|}{2} \right)$$

where L_v is from Lemma 3.

Proof. Since $\dot{x} = -\alpha \mathcal{V}(x, y) / \|\mathcal{V}(x, y)\|$, we have $\dot{V}_1 = -\alpha \nabla F(x)^\top \mathcal{V}(x, y) / \|\mathcal{V}(x, y)\|$. Set $a = \|\nabla F(x)\|$ and $b = \|\mathcal{V}(x, y) - \nabla F(x)\| \leq L_v \|y - y^*(x)\|$ (Lemma 3). Then $\nabla F^\top \mathcal{V} \geq a^2 - ab$ and $\|\mathcal{V}\| \leq a + b$, so

$$\frac{\nabla F^\top \mathcal{V}}{\|\mathcal{V}\|} \geq \frac{a^2 - ab}{a + b} = a - \frac{2ab}{a + b} \geq \frac{a}{2} - \frac{b}{2},$$

where the last step follows from $(a - b)^2 \geq 0$. Substituting $b \leq L_v \|y - y^*(x)\|$ concludes the proof. $\square$

Lemma 8 (Time derivative of V_2). *Under the normalized dynamics (5), the lower-level Lyapunov candidate satisfies:*

$$\dot{V}_2 \leq L_g \alpha \|y - y^*(x)\| - \beta \|\nabla_y g(x, y)\|.$$

Proof. Differentiating V_2 along (5) gives $\dot{V}_2 = (\nabla_x g(x, y) - \nabla_x g(x, y^*(x)))^\top \dot{x} + \nabla_y g(x, y)^\top \dot{y}$. The first term is bounded by $L_g \|y - y^*(x)\| \|\dot{x}\| = \alpha L_g \|y - y^*(x)\|$, using Lipschitzness of $\nabla_x g$ (Assumption 4) and $\|\dot{x}\| = \alpha$ by normalization; the second equals $-\beta \|\nabla_y g(x, y)\|$ since $\dot{y} = -\beta \nabla_y g(x, y) / \|\nabla_y g(x, y)\|$. Combining yields the claim. $\square$

1) Convergence Analysis::

Theorem 2 (Finite Time Convergence). *Assume $f(x, y)$ and $g(x, \cdot)$ are strongly convex (Assumptions 1 and 2) and satisfy smoothness and boundedness Assumptions 4, 5, and 6. Let the system evolve according to the normalized gradient flow (5). If α, β are chosen such that:*

$$\beta \geq \frac{\alpha L_v}{2\mu_g} + \frac{\alpha L_g}{\mu_g},$$

then the composite Lyapunov function $V(t)$ satisfies $\dot{V} \leq -k\sqrt{V}$ for some $k > 0$. Consequently, $V(t) \rightarrow 0$ in finite time, implying $(x(t), y(t))$ converges to the global optimum $(x^, y^*(x^*))$ in finite time.*

Proof. From Lemmas 7 and 8:

$$\begin{aligned} \dot{V} = \dot{V}_1 + \dot{V}_2 \leq & -\alpha \left(\frac{\|\nabla F(x)\|}{2} - \frac{L_v \|y - y^*(x)\|}{2} \right) \\ & + L_g \alpha \|y - y^*(x)\| - \beta \|\nabla_y g(x, y)\|. \end{aligned}$$

Using quadratic growth (Lemma 4):

$$\begin{aligned} \|\nabla F(x)\| & \geq \sqrt{2\mu_F V_1}, \quad \|y - y^*(x)\| \leq \sqrt{\frac{2}{\mu_g} V_2}, \\ \|\nabla_y g(x, y)\| & \geq \sqrt{2\mu_g V_2}. \end{aligned}$$

Substituting:

$$\begin{aligned}\dot{V} \leq & -\frac{\alpha\sqrt{2\mu_F}}{2}\sqrt{V_1} \\ & + \left(\frac{\alpha L_v}{\sqrt{2\mu_g}} + \frac{L_g\alpha\sqrt{2}}{\sqrt{\mu_g}} - \beta\sqrt{2\mu_g} \right) \sqrt{V_2}.\end{aligned}$$

We define

$$\gamma_1 := \frac{\alpha\sqrt{2\mu_F}}{2}, \quad \gamma_2 := \frac{\alpha L_v}{\sqrt{2\mu_g}} + \frac{L_g\alpha\sqrt{2}}{\sqrt{\mu_g}} - \beta\sqrt{2\mu_g}.$$

Choosing β such that $\gamma_2 \leq 0$, i.e.,

$$\beta \geq \frac{\alpha L_v}{2\mu_g} + \frac{\alpha L_g}{\mu_g},$$

then $\dot{V} \leq -\gamma_1\sqrt{V_1} - |\gamma_2|\sqrt{V_2}$. Noting $\sqrt{V_1} + \sqrt{V_2} \geq \sqrt{V}$:

$$\dot{V} \leq -\min(\gamma_1, |\gamma_2|)(\sqrt{V_1} + \sqrt{V_2}) \leq -k\sqrt{V},$$

with $k = \min(\gamma_1, |\gamma_2|) > 0$. The differential inequality $\dot{V} \leq -k\sqrt{V}$ is a standard finite-time stability certificate [9]: integrating $\frac{d}{dt}\sqrt{V} \leq -\frac{k}{2}$ gives $\sqrt{V(t)} \leq \sqrt{V(0)} - \frac{k}{2}t$, so V vanishes in finite time $t \leq 2\sqrt{V(0)}/k$. $\square$

C. Case C: $f(x, y)$ satisfies PL Inequality with μ_f and $g(x, \cdot)$ μ_g -strongly convex

We relax the upper-level convexity to the PL inequality (Assumption 3), using the same normalized flow (5) as Case B. *Assumptions:* 1, 3, 4, 5, 6. *Dynamics:* Normalized flow (5) with $\mathcal{V}$ from (3).

1) *Key technical lemma: PL-type inequality for $\nabla F(x)$:*

Lemma 9 (PL-type inequality for $F(x)$). *Suppose $f(x, y)$ satisfies the PL condition (Assumption 3) and $g(x, y)$ is strongly convex. Then the implicit hypergradient $\nabla F(x)$ satisfies:*

$$\|\nabla F(x)\|^2 \geq \mu_f(F(x) - F^*) - C,$$

where $C = \max\left(0, -\mu_f(F^* - f^*) + \frac{1}{2}C_f^2 + \left(\frac{C_{gxy}C_f}{\mu_g}\right)^2\right)$ is a constant depending on the problem conditioning and the residual value of the inner problem, and $f^* := \inf_{x,y} f(x, y)$.

Proof. Setting

$$\begin{aligned}b &= \nabla_x f(x, y^*(x)), \\ c &= \nabla_{xy}^2 g(x, y^*(x))[\nabla_{yy}^2 g(x, y^*(x))]^{-1} \nabla_y f(x, y^*(x)).\end{aligned}$$

Then $\nabla F(x) = b - c$. Using $\|b - c\|^2 \geq \frac{1}{2}\|b\|^2 - \|c\|^2$ (Young's inequality). From the PL-Inequality on f substituting $y = y^*(x)$:

$$\frac{1}{2}\|\nabla f(x, y^*(x))\|^2 \geq \mu_f(f(x, y^*(x)) - f^*).$$

Since $\|\nabla f\|^2 = \|b\|^2 + \|\nabla_y f(x, y^*(x))\|^2$, we have

$$\frac{1}{2}\|b\|^2 \geq \mu_f(f(x, y^*(x)) - f^*) - \frac{1}{2}\|\nabla_y f(x, y^*(x))\|^2.$$

Also $\|c\|^2 \leq C_{gxy}^2 \mu_g^{-2} C_f^2$. Combining and using $F(x) = f(x, y^*(x))$:

$$\|\nabla F(x)\|^2 \geq \mu_f(F(x) - f^*) - \frac{1}{2}C_f^2 - \left(\frac{C_{gxy}C_f}{\mu_g}\right)^2.$$

Rearranging using $F(x) - f^* = (F(x) - F^*) + (F^* - f^*)$ gives the desired form with C as stated. **Note:** Since $F^* \geq f^*$, a looser but simpler bound is $C \leq \frac{1}{2}C_f^2 + \left(\frac{C_{gxy}C_f}{\mu_g}\right)^2$. $\square$

2) *Convergence analysis:* We use V_1, V_2, V from (7). Lemmas 7 and 8 remain as in Case B (same normalized flow and same proofs apply). Thus:

$$\begin{aligned}\dot{V} \leq & -\alpha\left(\frac{\|\nabla F(x)\|}{2} - \frac{L_v\|y - y^*(x)\|}{2}\right) \\ & + L_g\alpha\|y - y^*(x)\| - \beta\|\nabla_y g(x, y)\|.\end{aligned}$$

When $F(x) - F^* > \frac{C}{\mu_f}$ (outside the residual neighborhood), Lemma 9 implies

$$\|\nabla F(x)\| \geq \sqrt{\mu_f(F(x) - F^*) - C}.$$

Combining with the quadratic-growth bounds for the lower level yields:

$$\dot{V} \leq -\frac{\alpha}{2}\sqrt{\mu_f V_1 - C} + \left(\frac{\alpha L_v}{\sqrt{2\mu_g}} + \frac{L_g\alpha\sqrt{2}}{\sqrt{\mu_g}} - \beta\sqrt{2\mu_g}\right)\sqrt{V_2}.$$

Choosing β large enough so that the $\sqrt{V_2}$ coefficient is non-positive:

$$\beta \geq \frac{\alpha L_v}{2\mu_g} + \frac{\alpha L_g}{\mu_g}.$$

Then for $V_1 > \frac{C}{\mu_f}$ we have $\dot{V} < 0$, which drives the system into the residual neighborhood.

Theorem 3 (Neighborhood convergence). *Consider the bilevel problem under Assumptions 1, 3, 4, 5, and 6. Under the normalized gradient flow (5), with β chosen sufficiently large, the system converges to a bounded neighborhood of the optimal value:*

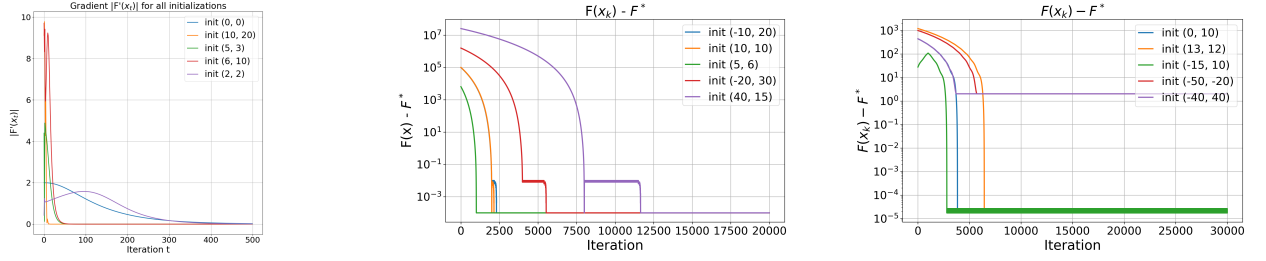
$$\limsup_{t \rightarrow \infty} F(x(t)) - F^* \leq \frac{C}{\mu_f}.$$

Proof. The analysis above establishes that $\dot{V} < 0$ whenever $V_1 > C/\mu_f$. The explicit constant C appears from Lemma 9. The parameter selection inequality ensures the V_2 coefficient is negative, and the $\sqrt{\cdot}$ term in V_1 drives a decrease until the residual ball is reached. $\square$

3) *Interpretation:* The PL inequality on f alone does not guarantee global convergence: the composition $F(x) = f(x, y^*(x))$ accumulates a residual C from $\nabla_y f$ and the inter-level coupling, so the dynamics reach only a ball around F^* of radius $\propto C/\mu_f$.

V. IMPLEMENTATION AND RESULTS

We validate the analysis on synthetic bilevel problems spanning the three regularity regimes.



(a) Case A: $\|\nabla F(x)\|$ decay

(b) Case B: finite-time convergence

(c) Case C: neighborhood convergence

Fig. 1. Numerical validation across the three regimes: (a) Case A, asymptotic gradient vanishing (Theorem 1); (b) Case B, finite-time convergence (Theorem 2); (c) Case C, convergence to a residual neighborhood (Theorem 3).

A. Discretization and Setup

The continuous-time dynamics are simulated using a Forward Euler discretization scheme. For a time step δ_t , the update rules are:

$$\begin{aligned} x_{k+1} &= x_k - \delta_t \alpha \cdot \mathcal{D}_x(x_k, y_k), \\ y_{k+1} &= y_k - \delta_t \beta \cdot \mathcal{D}_y(x_k, y_k), \end{aligned}$$

where $\mathcal{D}_x, \mathcal{D}_y$ correspond to either (5) or (6) depending on the regime. Unless otherwise stated, we use $\alpha = 0.1$, $\beta = 0.5$, and $\delta_t = 0.05$. We initialize from multiple starting points to demonstrate robustness.

B. Example 1: Non-Convex Upper Level (Case A)

With $f(x, y) = \sin(x) + \sin(y) + \sin(x+y)$ and $g(x, y) = 10y^2 - 2y + yx^3$, we employ the standard flow (6). As global optimality is not guaranteed, we track gradient decay. Fig. 1(a) shows the hypergradient norm $\|\nabla F(x)\|$ vanishing asymptotically, confirming Theorem 1.

C. Example 2: Strongly Convex Bilevel Optimization (Case B)

With $f(x, y) = 4x^2 + xy + 10y^2$ and $g(x, y) = (y - x^2)^2 - 20$, we apply the normalized flow (5) to test finite-time convergence. Fig. 1(b) shows the error reaching machine precision in finite time, confirming the $\dot{V} \leq -k\sqrt{V}$ collapse to the unique optimizer $(x^*, y^*(x^*))$ per Theorem 2.

D. Example 3: Upper-Level PL Condition (Case C)

With $f(x, y) = x^2 + y^2 + 3\sin^2(x)$ and $g(x, y) = (x - y)^2 + 2(y - 100)^2$, f satisfies the PL condition and g is strongly convex. We apply the normalized flow (5). As Fig. 1(c) shows, the iterates converge not to a point but to a bounded neighborhood of the optimum, validating Theorem 3.

VI. CONCLUSION

We presented a continuous-time, Lyapunov-based analysis of gradient-flow dynamics for bilevel optimization. The proposed normalized flow achieves finite-time convergence to the global optimum when both levels are strongly convex (stronger than the asymptotic or linear rates of discrete-time methods) while relaxations yield asymptotic convergence for non-convex upper levels and convergence to a bounded residual neighborhood under a PL upper level.

REFERENCES

- [1] L. Franceschi, P. Frasconi, S. Salzo, R. Grazzi, and M. Pontil, “Bilevel programming for hyperparameter optimization and meta-learning,” in *International conference on machine learning*. PMLR, 2018, pp. 1568–1577.
- [2] J. F. Bard and J. T. Moore, “A branch and bound algorithm for the bilevel programming problem,” *SIAM J. Sci. Comput.*, vol. 11, pp. 281–292, 1990.
- [3] C. N. Jones and M. Morari, “Approximate explicit mpc using bilevel optimization,” *2009 European Control Conference (ECC)*, pp. 2396–2401, 2009.
- [4] S. Ghadimi and M. Wang, “Approximation methods for bilevel programming,” *arXiv preprint arXiv:1802.02246*, 2018.
- [5] F. Pedregosa, “Hyperparameter optimization with approximate gradient,” in *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016, pp. 737–746.
- [6] D. Maclaurin, D. Duvenaud, and R. P. Adams, “Gradient-based hyperparameter optimization through reversible learning,” in *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 2015, pp. 2113–2122.
- [7] S. Dempe and J. Dutta, “Is bilevel programming a special case of a mathematical program with complementarity constraints?” *Mathematical Programming*, vol. 131, no. 1, pp. 37–48, Feb. 2012.
- [8] M. Arbel and J. Mairal, “Non-convex bilevel games with critical point selection maps,” *NeurIPS 2022 (arXiv preprint arXiv:2207.04888)*, 2022.
- [9] K. Garg, M. Baranwal, and D. Panagou, “A fixed-time convergent distributed algorithm for strongly convex functions in a time-varying network,” in *2020 59th IEEE Conference on Decision and Control (CDC)*, 2020, pp. 4405–4410.
- [10] H. Shen and T. Chen, “On penalty-based bilevel gradient descent method,” in *International Conference on Machine Learning (ICML)*, vol. 202. PMLR, 2023, pp. 30992–31015.
- [11] T. Chen, Y. Sun, Q. Xiao, and W. Yin, “A single-timescale method for stochastic bilevel optimization,” in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, vol. 151. PMLR, 2022, pp. 2466–2488.
- [12] R. I. Boj, E. Chenchene, E. R. Csetnek, and D. A. Hulett, “Accelerating diagonal methods for bilevel optimization: Unified convergence via continuous-time dynamics,” 2025.
- [13] Q. Xiao, S. Lu, and T. Chen, “A generalized alternating method for bilevel optimization under the polyak-łojasiewicz condition,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [14] S. Sharifi, N. Abolfazli, E. Y. Hamedani, and M. Fazlyab, “Safe gradient flow for bilevel optimization,” in *2025 American Control Conference (ACC)*, 2025, pp. 1675–1680.
- [15] H. Karimi, J. Nutini, and M. Schmidt, “Linear convergence of gradient and proximal-gradient methods under the polyak-łojasiewicz condition,” in *Joint European conference on machine learning and knowledge discovery in databases*. Springer, 2016, pp. 795–811.